



Mr. SOK PHANIT

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EDUCATION

2023.09 – 2023.08

Gyeongsang National University, South Korea

M.S. in Mechatronic Engineering

- GPA 4.43 / 4.5
- Thesis title: “Analysis and Implementation of the Crowbar Clamping Circuit for Surge Voltage Reduction in the DC Solid-State Circuit Breaker”. *IEEE SCI(E) journal publication recognized for graduation.*

2018.06 – 2022.04

Rajamangala University of Technology Thanyaburi, Thailand

B.S. in Mechatronic Engineering

- GPA 3.88 / 4.0 [First-Class Honors Graduate]
- Thesis title: “Automatic Transport Vehicle Indoor”

SKILLS AND RESEARCH INTERESTS

- Electronics & Electrical circuits simulation using PSIM, TINA-TI, PSpice, LTspice.
- Design of PCB using Altium Designer
- Deep learning (AI)
- Protection function: OCP, OVP, DESAT & Short-Circuit in EV inverter
- Converter system [AC/DC, DC/AC, DC/DC]
- EV inverters and motor drivers.
- Reliability analysis of the electronic circuit
- MCU, Python, C, DSP programming
- Data acquisition device and data analysis

PERFORMED PROJECTS DURING B.S, M.S & CURRENTLY WORK

DC solid-state circuit breaker (M.S)

- Design and performance of a voltage clamping circuit for a semiconductor DC solid-state circuit breaker.

Reliability analysis of hybrid power conversion device for household energy management(M.S)

- Development of a 220 VAC inverter for household use, designed to supplement grid power with a 5 kW capacity using various clean energy sources such as PV, ESS, and PEMFC to ensure high reliability.

Development of AI-based life prediction using the LSTM algorithm (AI) (M.S)

- This project can predict lifespan in real time, regardless of sensor failure, and secure high reliability of the Advanced Air Vehicle (AAV) power management system.

Reliability Analysis of PFC Converter in 4K TVs using Markov Approaches (M.S)

- To analyze an actual board to determine how faults in the PFC feedback signal section affect the converter's failure rate and lifespan, and to propose enhancing reliability in real-world TV applications.

AI research specializing in deep learning models for computer vision (B.S)

- This research aimed at enhancing the frame rate (FPS) performance of image classification tasks on the NVIDIA Jetson Nano single-board computer. More details [<https://github.com/SokPhanith>].

150 kW Next-Generation EV Inverter (Currently work)

- Developed a 150 kW EV inverter from scratch with H/W and S/W, featuring advanced protection, diagnostics, low-loss SiC switching, and robust operation in harsh environments.

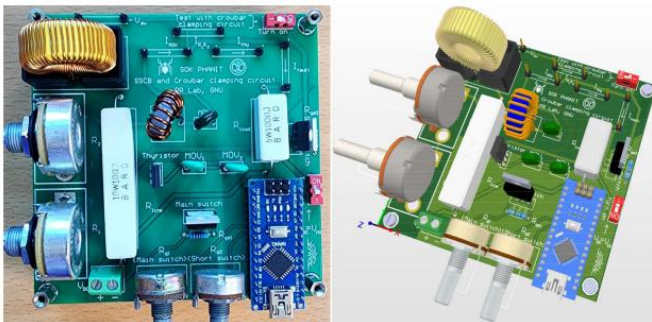
PATENT

[1] **P. Sok**, and F. -S. Kang “The crowbar clamping circuit to reduce the surge voltage in the DC solid-state circuit breaker” *Patent No. KR102810660B1*, May 2025. [Published]
[<https://patents.google.com/patent/KR102810660B1/en>]

[2] **P. Sok**, and F. -S. Kang “Triple-MOV-C snubber circuit for reducing surge voltage of DC circuit breaker” *Patent 2025-1-101-KR*, 2025. [Pending]

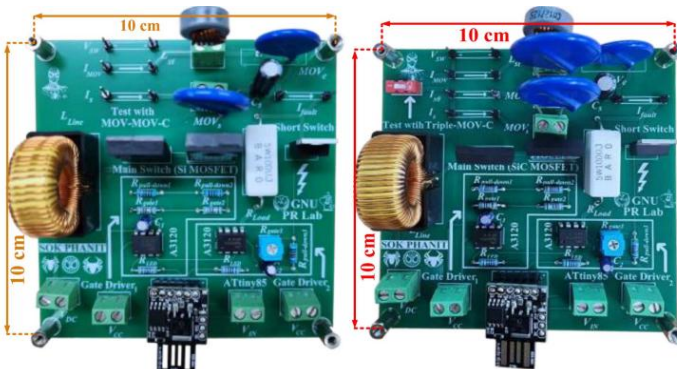
INTERNATIONAL JOURNALS

[1] **P. Sok**, S. -G. Song, and F. -S. Kang, “Analysis and Implementation of the Crowbar Clamping Circuit for Surge Voltage Reduction in the DC Solid-State Circuit Breaker,” *IEEE Access*, vol. 12, pp. 149018-149031, 2024.



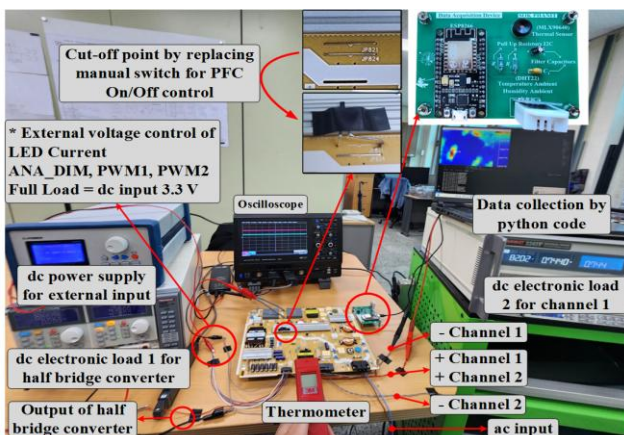
This PCB was designed by me in Altium Designer for testing in the journal. [SCI(E), Q1]

[2] **P. Sok**, S. -G. Song, G. Zhou, and F. -S. Kang, “Improving Surge Voltage and Power Loss of the MOV-MOV-C Approach for Low-Scale DC Solid-State Circuit Breaker Applications,” *IEEE Open Journal of the Industrial Electronics Society*, vol. 6, pp. 708-721, 2025.



This PCB was designed by me in Altium Designer for testing in the journal. [SCI(E), Q1]

[3] **P. Sok**, S. -Y. Joo and F. -S. Kang, "Markov Model-Based Reliability Analysis of LED TVs on PFC Converters With Control Loop Effects," *IEEE Transactions on Consumer Electronics*, vol. 72, no. 1, pp. 51-64, Feb. 2026.



Experiment configuration in this journal.

INTERNATIONAL AND DOMESTIC CONFERENCES

- [1] **P. Sok, S. -P. Kim, S. -G. Song, and F. -s. Kang**, “The Crowbar Clamping Circuit to Reduce the Surge Voltage in the DC Solid-State Circuit Breaker,” *10th International Conference on Applied System Innovation (ICASI)*, Kyoto, Japan, 2024, pp. 297-299.
- [2] **P. Sok, S. -P. Kim, F. -S. Kang, and S. -J. Park**, “Reliability analysis of hybrid power conversion systems by a fault tree analysis approach,” *Power Electronics Society Conference Proceedings (KIPE)*, Jeju-do, South Korea, 2024, pp. 663-664.
- [3] **P. Sok, S. -G. Song, and F. -S. Kang**, “Enhancing the MOV-MOV-C Approach to Reduce Power Loss and Surge Voltage in Low-Scale DC Solid-State Circuit Breakers,” *Spring Conference of the Electrical Equipment and Energy Conversion System Division (KIEE)*, Gangwon-do, South Korea, 2025, pp. 163-166.

AWARDS

31st Young Pioneer Researcher Award	Gyeongsang National University, South Korea <i>This ceremony presents the award for the graduate student who published an excellent article in a SCI(E) journal in 2025.05.</i>
Best Conference Paper Award	Korean Institute of Electrical Engineers (KIEE), South Korea <i>This award was provided based on my presentation conference paper, and it was the best paper at a conference in 2025.05.</i>
Award for Excellent Academic Achievement	National Institute for International Education (NIIED), South Korea <i>This award was granted based on my SCI(E) journal, which was recognized as one of the best among all GKS scholars, awarded in 2024.12.</i>
Grade A, Cambodia National Exam (Diploma)	Ministry of Education, Youth and Sport, Cambodia <i>The national diploma certificate exam was administered across Cambodia, and I was one of the top 405 Grade A students nationwide in 2016.09.</i>
Top 7th of Outstanding physics student	Ministry of Education, Youth and Sport, Cambodia <i>I took the national exam to find outstanding students around Cambodia, and I was placed in the top 7th nationwide among outstanding physics students in 2016.05.</i>

PROFESSIONAL EXPERIENCE AND VOLUNTEER WORK

Internship program at Survitech Co., Ltd, Thailand (2021.07-11)

- The work involved developing and designing a PCB prototype using Altium Designer software, coding a touchscreen HMI, and installing a CNC plasma system.

A full-time researcher at Power Electronics and Reliability, Gyeongsang National University Jinju, South Korea, (2023.08-2025.08)

- Design a DC solid-state circuit breaker, reliability analysis of various converter systems, DSP coding, and design a PCB prototype using Altium Designer.

H/W Design and Testing of the EV Inverter system, Hyowon PowerTech, Busan, South Korea (2025.10-present)

- R&D of HV battery systems for EV inverters and motor drivers, Responsible: covering circuit simulation (PSIM, TINA-TI, PSpice, LTspice), H/W and PCB design (Altium Designer), Function Protection and testing to validation, debugging, and design optimization using various engineering tools.

A volunteer's activities at the Cambodian Student Association in South Korea.

- We prepare cultural performances and various activity showcases to share in South Korea.
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TECHNICAL SKILLS

SCORE

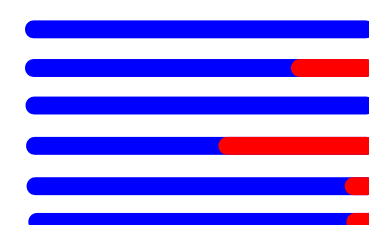
Languages

- Khmer - Native language.
- Thai - fluent [all aspects].
- English - fluent [all aspects].
- Lao - fluent in communication [Speaking and Listening].
- Korean - Intermediate [TOPIK II, Level 3].



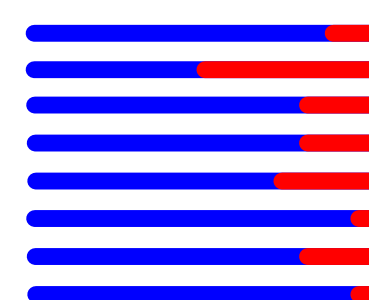
Experimental Equipment

- HV/LV of AC/DC source.
- Electronic Load, Power Analyzer, and Pulse Generator.
- Digital and analog multimeter.
- Electrical panel wiring [HV/LV/PLC systems].
- Oscilloscope.
- Thermal Detection & Recorded Thermal Analysis
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Software Tools

- PSIM, TINA-TI, PSpice, LTspice for electronic circuit simulation
- Code Composer Studio for DSP coding.
- Python code for deep learning [AI].
- Programming MCU.
- 3D design in SolidWorks.
- PCB design in Altium Designer.
- Linux operating systems of Raspberry Pi & Jetson Nano.
- General Computer skills and Microsoft Office.



SELF-INTRODUCTION

As a child born in Cambodia, I developed a strong interest in technology, inspired by a neighbor who taught me about computers and various electronic devices that he had begun to engineer. These experiences sparked curiosity and deepened my passion for technology. As a child, I dreamed of becoming the best engineer, particularly drawn to robotics and technology. I put in my best effort throughout high school and was honored to be one of Cambodia's top 7th outstanding physics students, and received a grade A on my diploma. Continued my journey at university and chose a major that aligned with robotics, circuits, and technology. Thus, my major is Mechatronic Engineering. Both my bachelor's and master's degrees were awarded full support by scholarships in Thailand and South Korea. Despite the challenge of not speaking the local language but I overcame the barriers and graduated with first-class honors [GPA 3.88/4] for my B.S. degree and am expected to graduate with an excellent thesis based on my IEEE SCI(E) journal for M.S. degrees [GPA 4.42/5]. During being a researcher, I worked on an SCI(E) journal, a patent, and several conference papers, and I also received multiple awards, as referenced in those research papers. I developed a problem-solving mindset, always eager to find solutions even in the most challenging situations. This positive attitude became my greatest strength and guided me throughout my academic journey. Language was a challenge, like Korean and English, while living abroad, but I always reminded myself that "It takes more courage to face our weaknesses than to seek escape."

I possess hard skills in areas related to my major, particularly in designing PCB prototypes Altium Designer, with strong knowledge of electronic devices. For circuit simulation, I primarily used PSIM TINA-TI, PSpice, LTspice for analysis. I also have experience programming various microcontroller boards using Arduino IDE, Micro Python, C, and including a DSP. Having a good understanding of Python code for a range of libraries and having strong experience in AI focus on computer vision, using deep learning for image classification and object detection. I also have a bit of experience using SolidWorks, as well. I also gained hands-on experience with electrical panel wiring in HV/LV/PLC systems.
